
Netaji Subhas University of Technology

Formerly NSIT, University of Delhi

Department of Computer Science & Engineering

Data Visualizations using Power BI



MINI PROJECT REPORT

Unit III — Automation and Data Visualization

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GitHub Repository

github.com/TanishyadavNSUT/Data-Visualizations-using-PowerBI

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1 Introduction and Dataset Overview

This project aims to explore, analyze, and communicate complex underlying patterns within acoustic data through effective data visualization techniques utilizing **Microsoft Power BI**. By integrating various multi-dimensional data plots, statistical distribution charts, and summary reports, we provide an intuitive look into feature extraction characteristics like MFCC (Mel-frequency cepstral coefficients) profiles, model accuracy metrics, and dataset composition.

The dataset used for this project involves high-dimensional audio feature vectors (AeroGuard dataset), intended for classifying complex sound signatures. Understanding this data visually is critical before feeding it into predictive models. Power BI provides the robust environment needed to generate these dynamic exploratory visualizations.

2 Distribution and Feature Variability

2.1 MFCC Boxplot

Visualization Concept. A Box Plot (or whisker plot) displays the five-number summary of a set of data: the minimum, first quartile, median, third quartile, and maximum. It is highly effective for identifying outliers and understanding the variance in audio feature sets.



Figure 1: Distribution of MFCC features using Boxplots

Insight. The boxplot reveals the central tendencies and the spread of MFCC components across multiple audio samples. High variability in specific coefficients indicates active frequency bands.

2.2 MFCC Violin Plot

Data Application. Unlike basic box plots, violin plots feature a kernel density estimation of the underlying distribution, enabling the viewer to perceive the distribution shape and multimode structures simultaneously.



Figure 2: Violin Representation of MFCC standard deviations

2.3 Feature Variability and MFCC Profile

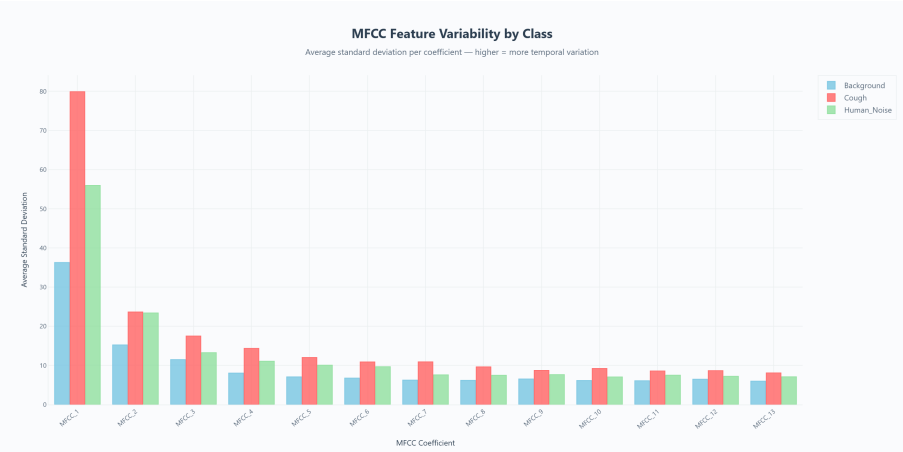


Figure 3: Feature Variability Plot across different sample components

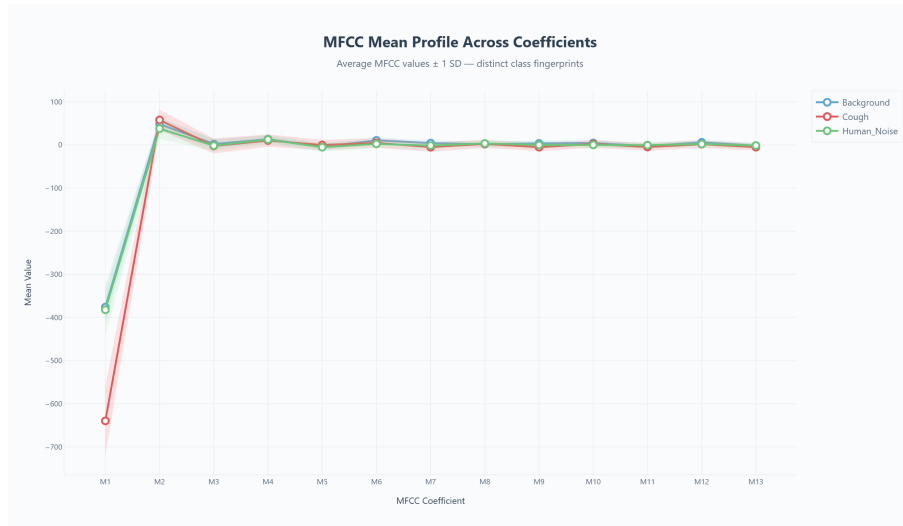


Figure 4: Average MFCC Profile curves revealing standard feature shapes

Insight. The feature variability plot identifies which specific components have the largest dynamic ranges, while the profile traces the standard shape of the MFCC vector across different audio classes.

3 Density and Component Spreads

3.1 Window Density Visualization

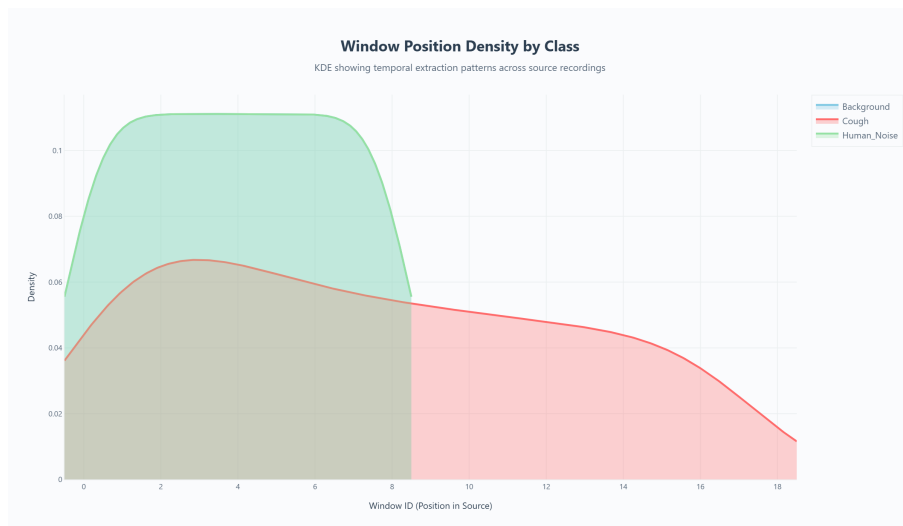


Figure 5: Density map showing the distribution across sliding windows

Visualization Concept. Density plots visualize the concentration of data points over a continuous interval or time period. It smooths out the noise to show regions of dense sample grouping.

3.2 Basic MFCC Scatter

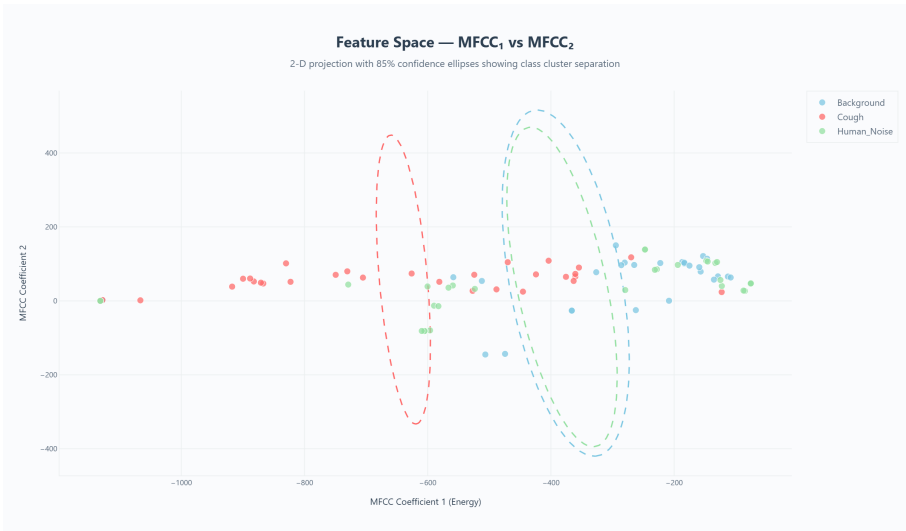


Figure 6: 2D Scatter plot indicating early separation of classification clusters

4 Advanced Correlation & Spatial Mapping

4.1 Correlation Heatmap

Visualization Concept. A correlation heatmap visually quantifies how strongly pairs of variables linearly relate to one another using a color gradient scale.

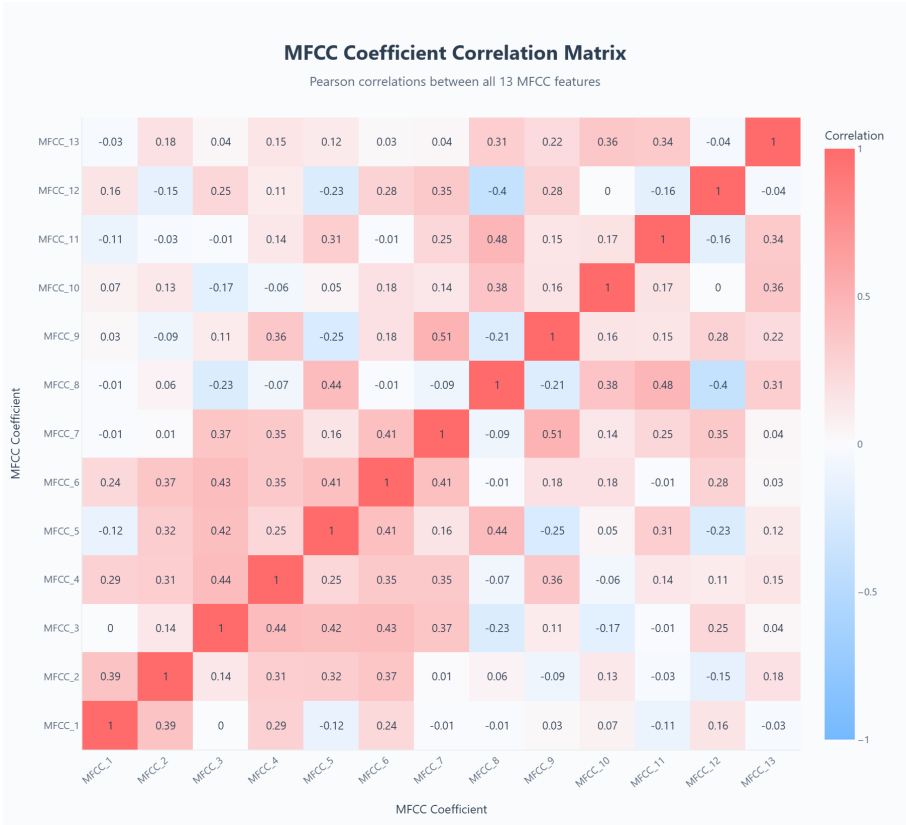


Figure 7: MFCC Correlation Heatmap indicating feature interdependency

4.2 Scatter Matrix

Data Application. A scatter matrix plot combines multiple bivariate scatterplots into a grid configuration. This is crucial for inspecting pair-wise relationships comprehensively across many dimensions.



Figure 8: Comprehensive Scatter Matrix for selected principal components

4.3 Multi-dimensional Geometries

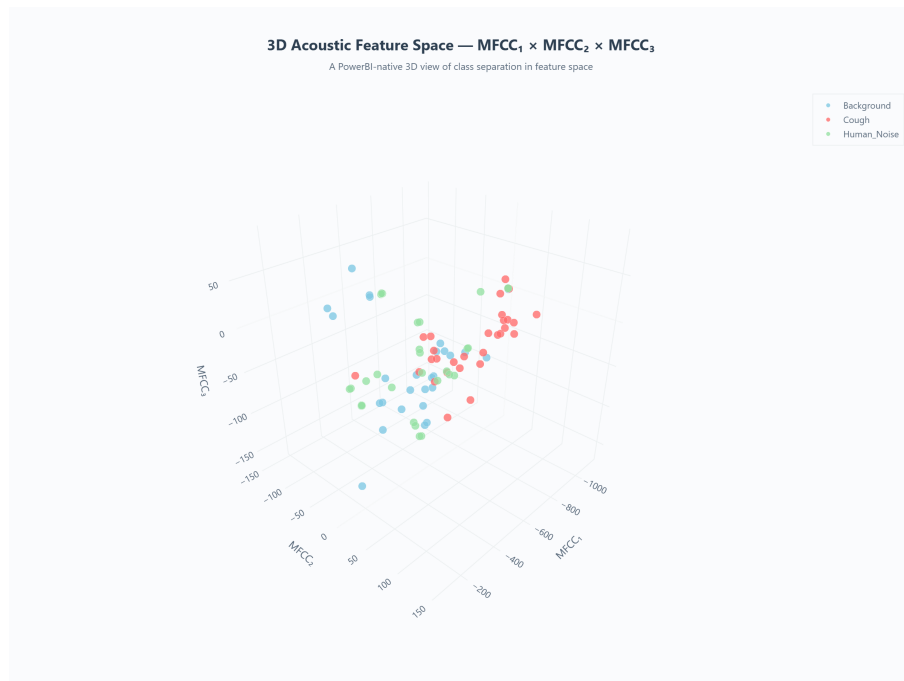


Figure 9: 3D Scatter Mapping showing spatial clustering of features

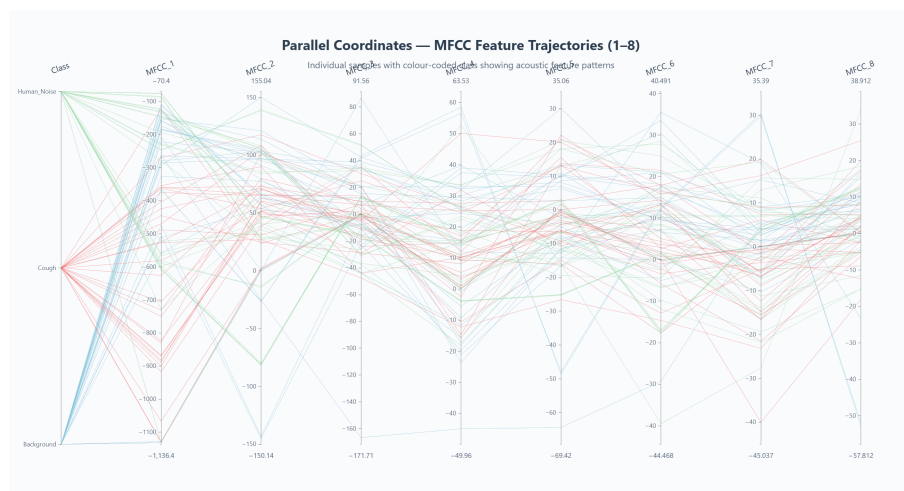


Figure 10: Parallel Coordinates Plot mapping high-dimensional trajectories

Insight. The 3D scatter provides spatial clustering based on three principal features. The Parallel Coordinates plot maps individual high-dimensional data points across vertical axes, allowing us to see trajectory patterns and feature thresholds for specific classes.

5 Composition and Flow Diagnostics

5.1 Dataset Sunburst

Visualization Concept. Sunburst charts show hierarchical data spanning outwards from the root module, ideal for showing dataset breakdowns by class and subclasses.

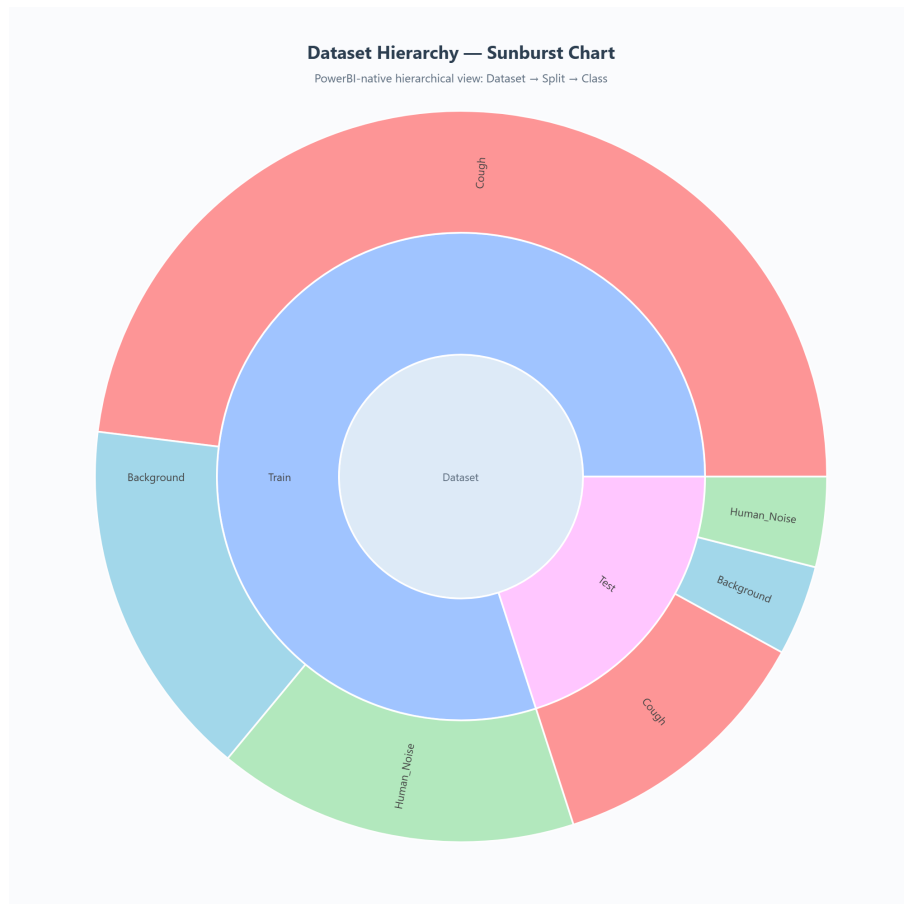


Figure 11: Sunburst Hierarchy of Dataset Labels

5.2 Dataset Flow and Cascades

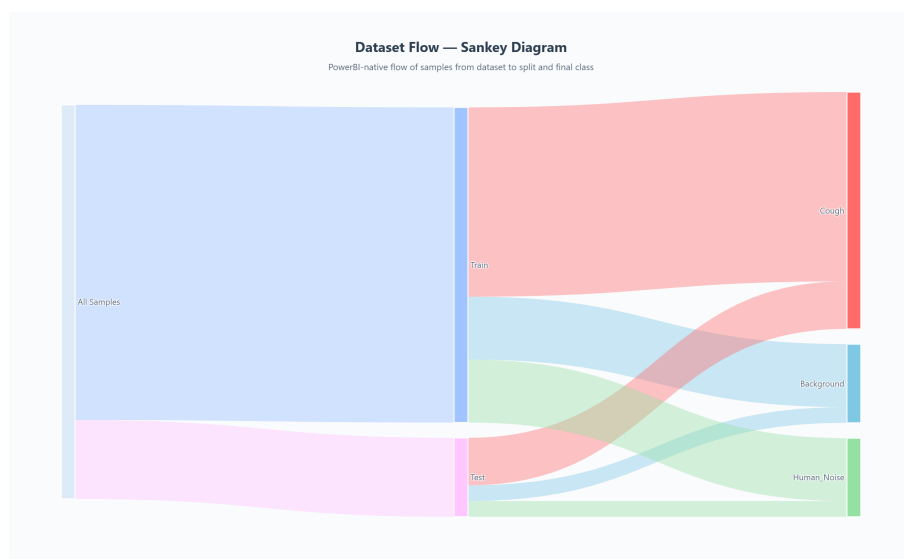


Figure 12: Sankey Diagram showing proportional data flow between preprocessing stages

5.3 Model Diagnostics: Funnel & Waterfall Charts

Data Application. Waterfall charts logically structure the cumulative effects of sequential values. The Funnel chart dictates progressive reductions in data logically passing from one testing phase to another.

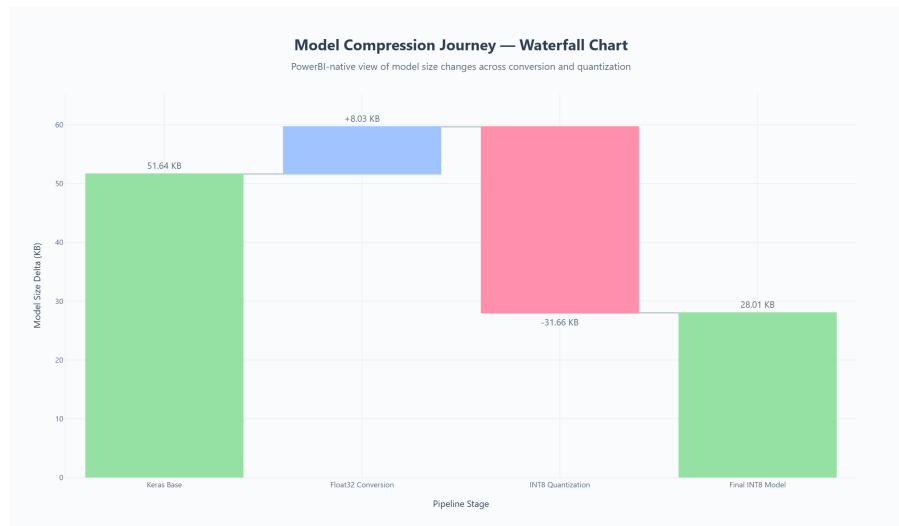


Figure 13: Waterfall Chart of Model Details and parameter composition

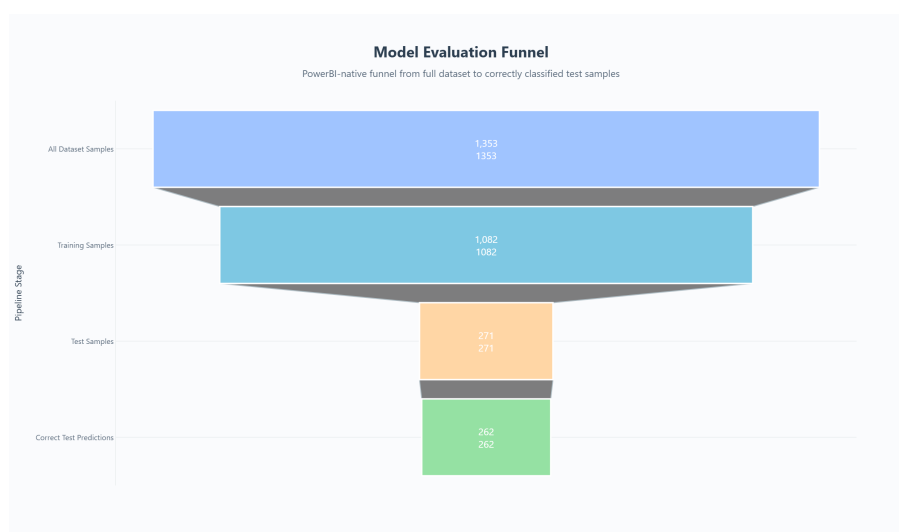


Figure 14: Data Funnel Pipeline demonstrating progressive filtering

6 Performance and Model Accuracy Metrics

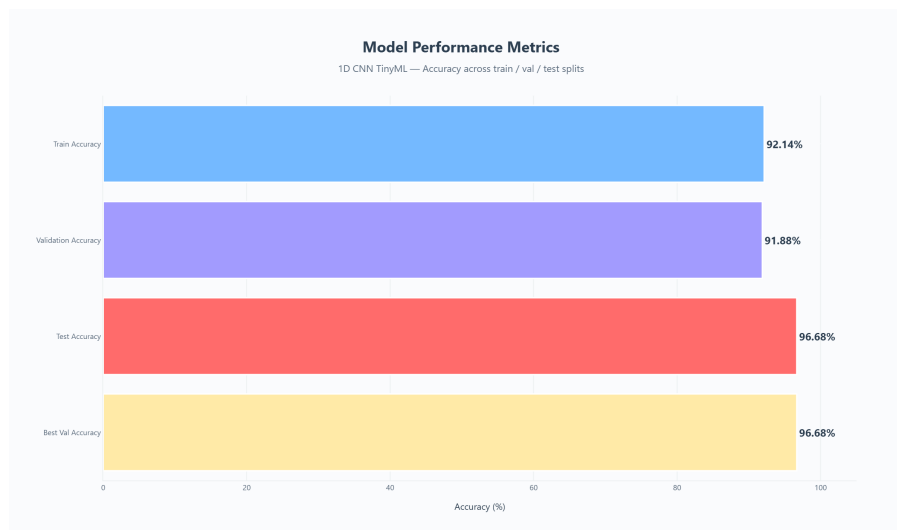


Figure 15: Epoch-wise Model Accuracy Progression Plot during training phases

Insight. The continuous tracking displays the learning curve of the underlying architecture, verifying steady convergence without pervasive overfitting.

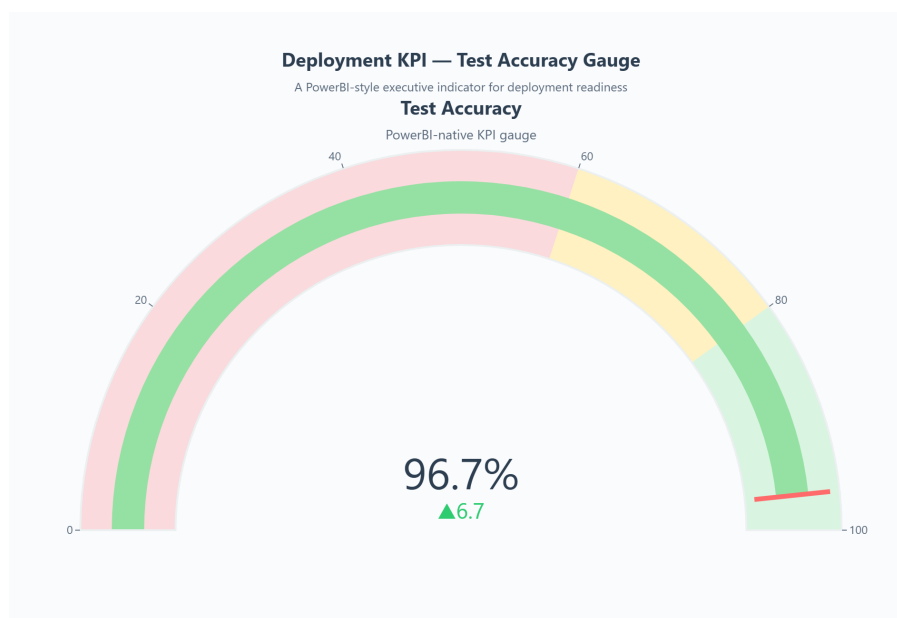


Figure 16: Final Test Accuracy Gauge Tracker

Insight. The standalone gauge metric distills the concluding accuracy into an easily readable KPI dashboard element, highlighting final benchmark success.

7 Comprehensive Executive Summary



Figure 17: Final Power BI comprehensive dashboard capturing all key insights.

Insight. This aggregated dashboard brings together descriptive, predictive, and interactive flow visualizations into a single view. The interconnected nature of the visuals within Power BI allows dynamic filtering based on specific sample classes, delivering a powerful executive reporting tool.